THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The shadow of a pole in the afternoon falls towards the: 1

(a) east **(b)** south only **(c)** north only **(d)** west

2. A pinhole camera cannot form an image if: 1

(a) the hole is small **(b)** the screen is translucent

(c) the hole is covered **(d)** the object is bright

3. A boy 1.2 m tall stands before a plane mirror. His image is: 1

(a) 0.6 m tall **(b)** 2.4 m tall **(c)** 1.2 m tall **(d)** 0 m

4. A lamp is moved away from an object. The shadow size: 1

(a) decreases **(b)** stays same **(c)** vanishes **(d)** increases

5. Which of these is formed on the screen of a pinhole camera? 1

(a) virtual image **(b)** real image **(c)** shadow **(d)** nothing

6. A person's image is 6 m from her. How far is she from the mirror? 1

(a) 6 m **(b)** 12 m **(c)** 3 m **(d)** 1.5 m

7. An object is moved towards the screen. The shadow size: 1

(a) vanishes **(b)** increases **(c)** decreases **(d)** stays same

8. The image of the Sun by a small gap in leaves on the ground is: 1

(a) no image **(b)** a shadow **(c)** a round spot **(d)** square

9. If a mirror is rotated by 10° keeping the incident ray fixed, the reflected ray turns by (basic idea): 1

(a) 20° **(b)** 10° **(c)** 0° **(d)** 5°

10. A small source of light produces: 1

(a) coloured shadow **(b)** blurred shadow **(c)** no shadow **(d)** sharp shadow

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How would you find the height of a tree using its shadow and a stick? 2

12. Why are the headlights in vehicles designed to avoid glare? 2

13. Explain why we can see a non-luminous object. 2

14. Why is the image in a plane mirror not formed on a screen? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how to make a pinhole camera and observe the image of a candle. 3

16. Explain the term 'image distance equal to object distance' in plane mirrors with an example. 3

17. Why are the mirrors in a car made slightly convex? (Hint: wider field of view) 3

OR

Explain why a shadow is formed on a sunny day but not on a cloudy day. 3

18. What is the use of two parallel mirrors in a barber shop? 3

OR

How does a kaleidoscope use reflections to form patterns? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the construction of a periscope and the path of light. 5

OR

Explain how reflection helps us see the Moon, planets and everyday objects. 5

20. Describe an experiment to verify that the angle of incidence equals the angle of reflection. 5

OR

Describe how a kaleidoscope is made at home and why patterns change. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND